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**ASTRONAUT PROFILE EVOLUTION THROUGH TIME AND SPACE: STUDY OF THE PAST,
CURRENT AND FUTURE REQUIREMENTS**

Tania Gres^{a*}, Erin Richardson^b, Megha Choudhary^c, Saira O. Williams^d, Luísa Santos^e, Marie Lambert^f

^a *Space Generation Advisory Council (SGAC), Italy, tania.gres@gmail.com*

^b *University of Colorado Boulder, United States, erin.richardson@colorado.edu*

^c *Space Generation Advisory Council (SGAC), India, meghachoudhary2509@gmail.com*

^d *Space Generation Advisory Council (SGAC), Costa Rica, roxysairawilliams@spacegeneration.org*

^e *Federal University of Rio Grande do Norte (UFRN), Brazil, luisasantos98@hotmail.com*

^f *Space Generation Advisory Council (SGAC), France, marie.lambert@spacegeneration.org*

* Corresponding Author

Abstract

Since 60 years ago when the first person was sent into space, humanity has seen around 630 astronauts leaving the Earth for orbital adventures of short or long duration. Nowadays, this dangerous and harsh environment attracts more and more people. It is well illustrated by the recent growing space tourism activity, and their new astronauts, that don't need to go through the classical recruitment and training as it was done for the past missions. We could categorise astronauts into 3 categories: the public and private astronauts whose profile evolved a lot through the years, and tourists astronauts which appeared very recently. When recruiting public astronauts today, they must have a completely new set of skills like interaction with robots, experience with new tools and technologies, and even being good public speakers. Private ones also must comply with standards imposed by national agencies and thus their recruitment could be a bit similar, which differs from tourist astronauts that are not expected to have these specific skills.

In the frame of the DIVERsity IN Astronaut Selection (DIVINAS) project which is part of the Diversity and Gender Equality Project Group from the Space Generation Advisory Council, a few members decided to take on the challenge of analysing astronauts' profile through the years and compare these data with recent crews. This paper aims to observe the evolution of the key parameters and capacities required through the years since the first human in space, in 1961. During the past 60 years, humanity went through different phases like the cold war with the space race, sustainability and collaboration in space with the space stations, as well as the return to the Moon and preparation of future long trips to Mars. Some criteria of selection like the number of years of experience, the background of the applicants or the physical skills they have will be studied, as well as the skills a "great astronaut" should have. This paper will also compare the recent recruitments, both in private and public agencies, as well as the new project of inclusivity like the parastronauts project of the European Space Agency. Thanks to the insights of recruiters in private and public agencies, as well as the data found online about past and current astronauts, this study will focus on highlighting the astronaut profile of tomorrow, and the future needs and skillset researched during recruitments, for the future of human spaceflights.

Keywords: astronaut, diversity, database, profile

Acronyms/Abbreviations

AEB	Brazilian Space Agency
ANGKASA	Angkatan Kerjasama Kebangsaan Malaysia Berhad
CNSA	China National Space Administration
ESA	European Space Agency
FAA	Federal Aviation Administration
ISA	International Space Agency
ISS	International Space Station
KARI	Korea Aerospace Research Institute
MBRSC	Mohammed Bin Rashid Space Centre
NASA	National Aeronautics and Space Administration
NOAA	National Oceanic & Atmospheric Administration
SGAC	Space Generation Advisory Council
STEAM	Science, Technology, Engineering, Arts and Mathematics
US	United States
VSS	Virgin Space Ship

twenty years in order to willingly highlight the improvements.

The objective of this work is to consider all individuals that ever went to space and display statistics showing the evolution of their profile and the reasons likely to explain these changes. With the start of space tourism and the recent increase in private missions, the last twenty years have witnessed a radical shift in the profile of new astronauts compared to the beginning of human space exploration.

Former papers on the subject such as “*A brief history of spaceflight from 1961 to 2020: An analysis of missions and astronaut demographics*” from Michael G. Smith, Michelle Kelley and Mathias Basner, cover the subject of astronaut profile evolution through time. With this paper, we considered a different axis of research. For example, the major differences between the two papers would be the criteria selected for study. While this paper considered the nationalities and ethnicities of every individual that ever went to space, the marital status and number of children are rather studied in the previously cited paper. These differences allow a different and complementary point of view and bring out diverse analyses.

This paper contains a detailed explanation of the astronaut database creation before covering a series of analyses on various subjects related to the data gathered. A first analysis on crew size, mission duration and type highlights the differences between the first twenty years and the last twenty years of human space exploration. The second analysis covers astronauts' backgrounds throughout the years. The third analysis focuses on private flights and astronauts, underscoring the diversity in gender, nationality and background. The last analysis studies the link between agencies and their number of astronauts, their gender, nationality and ethnicity, comparing the first twenty years and the last twenty years of human space exploration.

1. Introduction

This paper on the astronaut profile evolution through time and space aims at studying the past, current, and future requirements of space missions.

From the first human mission in space in the 1960s to today, the profile of astronauts selected to go to space has diversified and we documented new professional backgrounds, ethnicities, nationalities and an increasing number of women getting access to space.

Male military pilots and engineers were once the standard in astronaut recruitment. However, it is well documented that this criteria evolved through time. This paper aims at showing through data gathering the increase in diversity and mainly compares the data of the first twenty years of space exploration with the last

1.1. Timeline of Space Programs

1.1.1. Project Mercury

Project Mercury was the United States' first human space program. Over the course of six years, from 1958 to 1963, NASA performed six manned flights with the objectives of sending men to orbit in a spacecraft around Earth and returning them safely while studying their abilities to live in this new environment.

1.1.2. The Gemini Program

Gemini is NASA's following space program. For another six years, from 1961 to 1966, new missions were conducted to slowly train and prepare astronauts while studying the effects of long flights on the human organism. Equipment and spacecraft tests were also performed, which paved the way for spaceflight between the Earth and the Moon.

1.1.3. *The Apollo Program*

The Mercury and Gemini programs helped NASA prepare to send the first men to the Moon during the Apollo missions. From 1968 to 1972, the Apollo program allowed eleven space flights to be performed; the first four had the purpose of testing the equipment, while six of the seven following missions landed on the Moon. During these missions, twelve astronauts have walked on the lunar surface. They brought back lunar rocks, and these samples are still used today for study. [1]

1.1.4. *Almaz space stations*

The Almaz program was a top secret soviet military space station program that started in the early 1960s. Between 1973 and 1976, Salyut 2, Salyut 3 and Salyut 5 were launched and to cover the military nature of the program they were designated as Salyut civil space stations. After Salyut 5, in 1978, the Soviet Ministry of Defence judged that it was too expensive and outdated technology to still maintain its maintenance. [2]

1.1.5. *Skylab*

Skylab was the first American space station and was launched at an altitude of 435 km in 1973, but it did not last long due to its unplanned re-entry into the atmosphere in 1979. The name means laboratory in the sky due to the experimental purpose of research and training due to manned missions. [3]

1.1.6. *The Space shuttle era*

Recalling the Skylab period, it is noted that their development was essential for the existence of the International Space Station. In this period of vast innovation in the area of human flight technologies, the space shuttle was built, which carried people in orbit, recovered and repaired satellites, carried out highly complex research and made it possible to build the largest structure in space, the Space Station International. [1]

1.1.7. *The MIR era*

Mir was a space station operated by the Soviet Union and later by Russia, in low Earth orbit during the period 1986 to 2001, which was the first modular space station. At the time, it was mounted in orbit from 1986 to 1996 and was the largest artificial satellite in orbit, but with the fall of its orbit, there was the succession by the International Space Station (ISS). [4]

1.1.8. *The International Space Station era*

The international space station has flown 135 missions and is the largest modular space station currently in low Earth orbit. It is a multinational collaborative project involving five participating space agencies. Built between 1998 and 2011, the space station has been housing humans continuously since 2000. One of the big goals is to understand how molecules and cells behave differently in space. [5]

2. **Astronaut Database Creation**

We collected statistics on all humans ever launched to space through August 4, 2022. This includes publicly-funded astronauts, cosmonauts, and taikonauts hired by government agencies as well as people on privately funded missions. A spreadsheet was created with a row dedicated to each crew member per spaceflight, with people who have flown to space multiple times having one row per launch.

2.1. *Fields of interest*

The following information was recorded about each person at the time of launch, where available:

1. Name
2. Gender
3. Nationality
4. Second nationality (if applicable)
5. Ethnicity
6. Second ethnicity (if applicable)
7. Number of years of relevant experience
8. Occupation prior to selection
9. Sport played
10. Second sport played
11. Whether they have served in military
12. Whether they had pilot training prior to selection
13. Whether they had scuba diving training prior to selection
14. Space agency they belong to or Private
15. Crew size for the launch
16. Launch date
17. Launching country and whether the launcher is private or public
18. Mission type
19. Second mission type (if applicable)

20. Mission duration

2.2. Grouping of Answers

To simplify data collection and to enable comparison of statistics, possible answers for certain fields were grouped. Ethnicity groups were: American Indian or Alaska Native, Asian, Black or African American, Hispanic or Latino, Native Hawaiian or Other Pacific Islander and White, in accordance with the United State Office of Management and Budget 1997 Standards for Maintaining, Collecting, and Presenting Federal Data on Race and Ethnicity [6]. Occupations were also grouped into categories such as medicine, pilot, several engineering disciplines, several science disciplines, and others. Military experience, flying experience, and diving experience could be filled in with “Yes”, “No”, or “N/A” if information was not available. Launching country and type could be filled in with “American governmental”, “American private”, “Russian government”, or “Chinese governmental”. NASA’s recent missions on SpaceX’s Crew Dragon, for example, were labelled “American governmental” as they were launched from the United States but operated by a commercial provider. Mission types were grouped into: Firsts (early tests), satellite deployment, scientific experiment, station maintenance or deployment, commercial or advertisement, tracking and data relay satellite, earth observation, arts, lunar surface operations, telescope or spacelab deployment, telescope maintenance, and cargo resupply.

In total, there have been 1386 humans launched in space (including people who made several spaceflights), which makes 636 different people, on the date of the 05/09/2022.

3. Evolution of Crew Size and Mission Duration

We first investigate the evolution of crew size, mission duration, and mission type over time. The purpose of each mission as well as budget and mass constraints dictate the amount of people that can be and need to be launched. Government space agency budgets and space exploration objectives have changed over time, shaped by the political climate of each era.

3.1 The Early Days of Human Spaceflight (1961-1982)

The 1960s and 1970s saw rapid development in space activity. Characterised by the Cold War, the United States and Soviet Union were racing to prove their technological capabilities. This period served as a proving ground with missions aiming to achieve “firsts”, culminating in the first person on the moon followed by human exploration of the lunar surface.

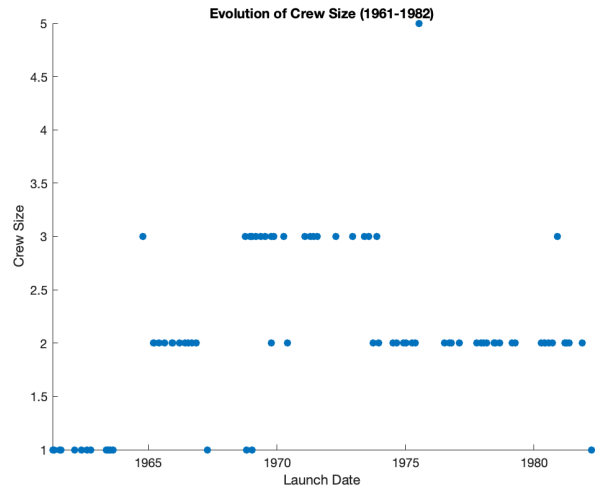


Figure 1: Launch crew sizes from 1961 until 1982.

In 1961, Yuri Garagin orbited the Earth and became the first person in space, launched by the Soviet Union. The United States’ first human spaceflight program, Project Mercury, aimed to put a human in orbit around the Earth, to study a person’s ability to function in space, and to safely recover the person and spacecraft [7]. These flights only required one astronaut and were also testing the impact of spaceflight on that astronaut before launching more people. Project Mercury ran until 1963, comprising some of the first group of data points in Figure 1.

In 1961, US President Kennedy set a goal for the United States to land people on the surface of the moon and return them safely by the end of the decade; the following spaceflights made progress towards achieving a crewed lunar landing [7]. With flights from 1964 to 1966, the Gemini Program prepared astronaut and ground crews for trips to the Moon by incorporating rendezvous and docking, improved reentry and landing methods, and longer missions [7]. The Gemini spacecraft carried two people, corresponding to the jump in crew size around 1964 in Figure 1. Both the Soviet Union and the United States’ early flights were short, with Yuri Gagarin and Alan Shepard’s first flights lasting less than two hours. Gemini aimed to test how astronauts could handle 14 day flights to prepare for trips to the Moon and thus required an increase in mission duration, as seen in Figure 2.

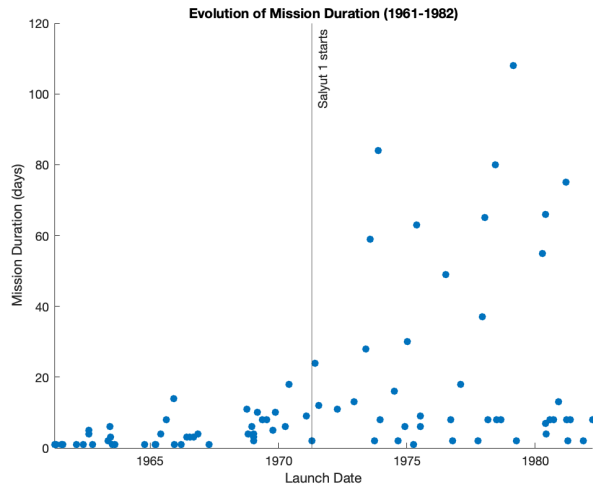


Figure 2: Astronaut mission durations from 1961 until 1982.

The Apollo Program saw crewed flights of 3 astronauts from 1968 to 1972. It consisted of test flights, lunar landings, scientific exploration of the moon, as well as building capability for human presence on the lunar surface [7]. Meanwhile in 1971, The Soviet Union launched the first space station, named Almaz, which saw mission durations increasing again, as shown in Figure 2, with the start of Almaz marked by the vertical line. Almaz's capacity for a rotating crew of three as well as Apollo's crew size of three are seen in the consistent data points in the early 1970s in Figure 1.

The United States' first space station, Skylab, set the foundation for the International Space Station (ISS) [7]. The three Skylab missions increased in length, lasting 28, 59, and 84 days [8]. These missions involved station set up and maintenance as well as science experiments and solar system observation studies [8].

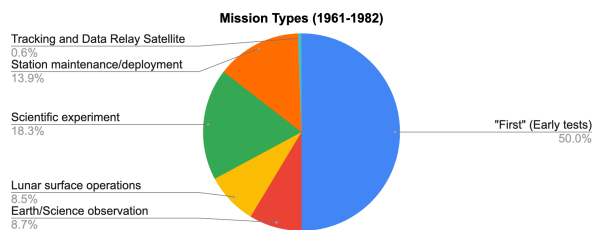


Figure 3: Make up of mission types from 1961 until 1982.

The objective of early human spaceflight missions influenced the amount of crew launched and how long they remained in space for. All launches from 1961 to 1982 assigned a mission type, with the results shown in Figure 3. Space "firsts" and early test flights comprise half of the flights, with other goals including lunar

surface operations, science experiments, and space station deployment and maintenance.

3.2 Contemporary Human Spaceflight (2002-2022)

Shifting to the 2000's, the goals of recent missions have drastically changed. Following Mir, The International Space Station provides a permanent fixture for habitation, a well known work environment, and environmental conditions for extensive microgravity experiments.

NASA's Space Shuttle program ran until 2011 and carried crews of 5 to 7 people. The Shuttle's retirement marks a large drop in crew size in 2011 where afterwards all launches relied on Russia's Soyuz spacecraft. Space Shuttle missions worked to build the International Space Station. Mission durations drastically increased as crew were able to stay on the ISS. The ISS has inhabited astronauts continuously since 2000 [7] with the record for longest time in space being broken multiple times..

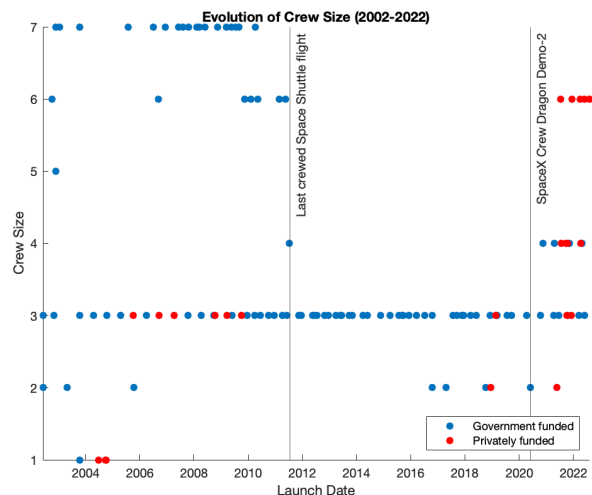


Figure 4: Launch crew sizes from 2002 until August 2022.

As shown in Figure 4, launch crew sizes remain consistently 3 or less people from the retirement of the Space Shuttle in 2011 until the first crewed flight of SpaceX's Crew Dragon in 2020. To demonstrate the advent and recent boom in privately funded spaceflight, Figures 4 and 5 mark government funded launches with blue and privately funded launches with red.

Since NASA began using Crew Dragon many government funded flights have a crew of 4 but Roscosmos' Soyuz remains in operation launching 3 people at a time.

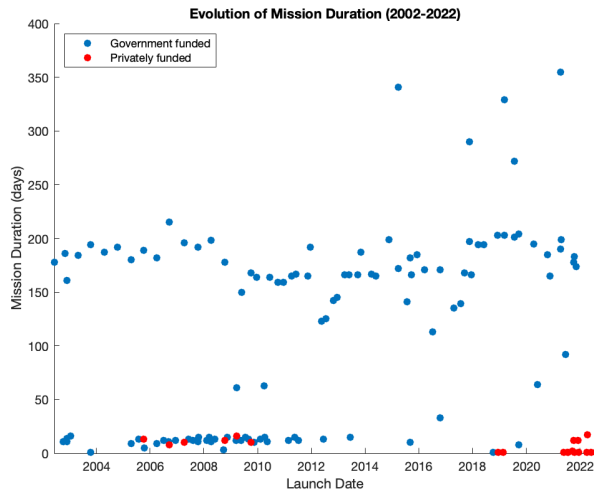


Figure 5: Astronaut mission durations from 2002 until August 2022.

Figure 4 shows recent privately funded space flights having higher crew sizes than government funded ones, as indicated by all launches with a crew size over four since the Space Shuttle era being privately funded. Space tourism companies have been launching customers in larger crew sizes; both Blue Origin and Virgin Galactic can carry six passengers at once.

Private flights are also shorter in duration, both lasting less than two hours. As shown in Figure 5, since 2018, most passengers flying less than 50 days in length are on privately funded missions. Both Blue Origin and Virgin Galactic's flights last less than two hours. Accommodating more passengers means more money can be made per flight, and tourism flights can provide the experience of visiting space in a short period without needing extended time for research like that conducted on the ISS.

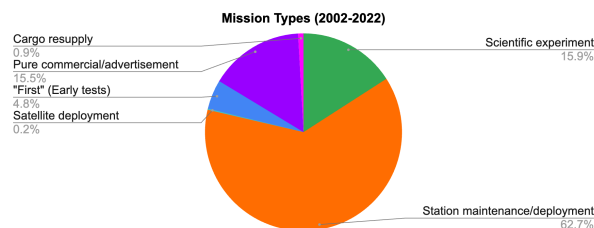


Figure 6: Make up of mission types from 2002 until 2022.

Like Figure 3, Figure 6 shows all launches from 2002 until 2022 grouped by mission type. Station deployment and maintenance missions form the majority, with scientific experiments and purely commercial (privately funded) missions following. Though this commercial category only forms 15% of this period's flights, this

number is significant considering the recency of these flights. As indicated by the red data points on Figures 4 and 5, we are experiencing a shift in how humans are getting to space. Considering private investment in the space industry and wealthy people's interests in experiencing spaceflight, this trend is likely to steepen. On the government side, space agencies around the world are setting goals of returning humans to the Moon and then to Mars. These missions will demonstrate new technology and operational concepts, requiring a shift back to the "first" early tests we observed in the Mercury, Gemini, and Apollo era.

4. Analysis of Astronaut Profile through the Perspective of their Professional Backgrounds

The early spaceflight programme called for highly trained military personnel - most of them people with flight experience. Early in the space program, military jet test piloting and engineering training were often necessarily required to be selected as an astronaut at NASA as can be seen in Fig 7 below. The earliest pool of astronauts were all white males with excellent training and physical fitness.

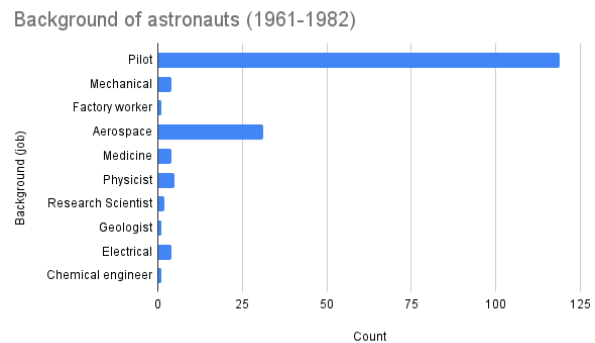


Figure 7: Background of astronauts from 1961-1982

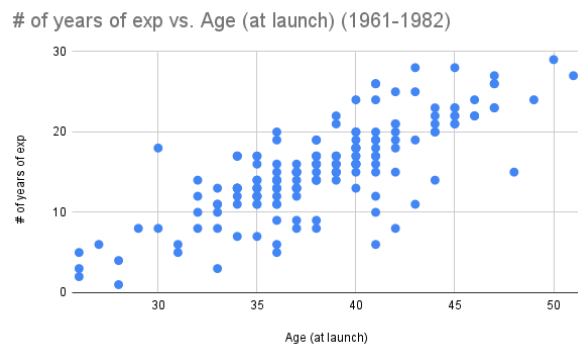


Figure 8: Number of years of experience v/s Age at Launch of astronauts from 1961-1982

Fig 8 shows the number of years of professional experience the astronauts had plotted against their age at launch. It can be seen that most of the astronauts fell between the age range of 32-45, with a majority of them having over 10 years of experience in their respective fields.

As can be seen from the above figure, almost 70% of the astronauts were pilots with engineers with different profiles comprising 23% of the astronauts.

Background (job)	Average Age at Launch
Aerospace engineer	38.38
Chemical engineer	39
Electrical engineering	40.25
Factory worker	26
Geologist	37
Mechanical engineer	41
Medicine	39.75
Physicist	39.6
Pilot	38.11
Research Scientist	37.5

Table 1: Background of astronaut with their average age at time of launch from 1961-1982

Almost all the astronauts from 1961-1982, no matter what their previous background was, were above the age of 37 at the time of their first launch.

With the access to space increasing rapidly over the past few decades, the space frontier is not just restricted to pilots and engineers.

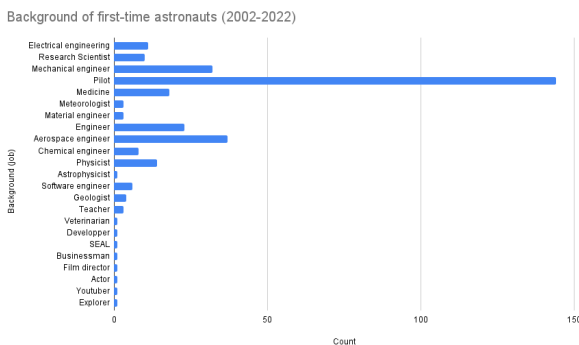


Figure 9: Background of first time astronauts from 2002-2022

Fig 9 demonstrates that even though pilots and engineers form a good majority of the astronauts, YouTubers, veterinarians, actors, film directors, etc can also now travel to space.

5. Space Tourists Profile Comparison

Since the first private astronaut in 2001, Dennis Tito, around 50 people have had the opportunity to fly in one of the most challenging places ever known. But who is considered as a space tourist exactly? Where are they from and what is their profile? Let's discover how the private industry and private space flights increased diversity on their own.

5.1. Introduction to space tourists: What/Who are they?

First of all, there are several definitions existing of what makes a space tourist. These definitions seem to be based on the purpose of their flight, their destination, as well as the origin of their funding.

About the altitude: Different entities consider the starting of space at different altitudes:

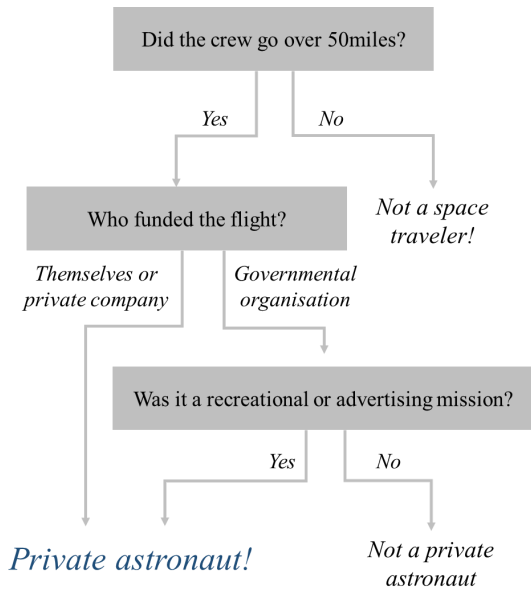
- For the entities who use the Kármán line, the limit of space is 62 miles up, or roughly a hundred kilometres above sea level.
- For the Federation Aeronautique Internationale (FAI), Space begins a hundred kilometres up too.
- The Federal Aviation Administration (FAA), the U.S. Air Force, the National Oceanic & Atmospheric Administration (NOAA), and NASA generally use 50 miles (around 80 kilometres) as the boundary with the Air Force granting astronaut wings to flyers who go higher than this mark.

In the frame of this paper, it is chosen to set at 50 miles (roughly 80km) the beginning of space, which makes people flying over it considered as space travellers.

About the funding and the spaceflight purpose: Has been considered as space tourists the people who went in space self-funded or funded by a private organisation. But in the case they are funded by a governmental institution, tourists are the ones who conducted a mission with a recreational or advertising purpose.

The definition of "recreational purpose" in space is in fact losing its sense with the new use of space, and the integration of new industries to it (like the food industry, cinema industry, music industry, arts in general...). The disruptive activities in space as of today will most likely become the norm for tomorrow, so this analysis simply decided to include in "recreational and advertising purpose" anything which is different from medical and science experiments, maintenance activities, space systems deployments or refurbishments missions.

The following chart recaps the above paragraph:



In terms of number and repartition over time, we can see a clear rise in private astronaut numbers in the next chart. Between 2001 and the end of 2009, private astronaut flights were quite regular, with roughly one launch per year. At this time, there were approximately 30 people launched in space per year (private and non private launches included), which makes a percentage of 3% of private flights over the total number. Since the Virgin Galactic's flight in 2018 which announces the return of private space flights, half of the people in space were private (56 private flights over 112 flights). We also noticed an increase in the total number of space flights since 2021, when *Blue Origin* and its *New Shepard*, as well as *Virgin Galactic* with its *VSS Unity* have visited space, with quite big crews onboard.

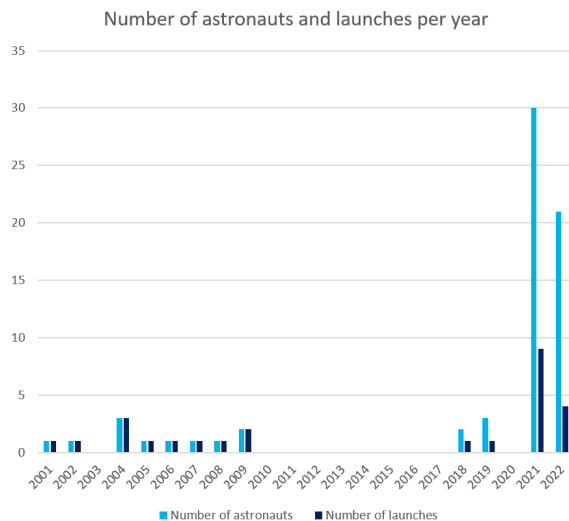


Figure 10: Number of private astronauts and launches per year

Between 2001 and 2009, *Space Adventures* was the only organisation to provide access to space to private astronauts, through the Soyuz capsule. They also sent the first South-African and first Iranian in space, while sending people from various nationalities as well.

5.2. Geographical representativity

Here is a list of all the different nationalities of people to have ever launched in space:

Table 2: Numbers of each astronauts' nationality that performed private space flights

Nationality	Number
United States	47
United Kingdom	8
Canada	3
South Africa	3
Japan	2
Russia and the Soviet Union	2
Hungary	2
India	1
Netherlands	1
Iran	1
Australia	1
Israel	1
Brazil	1
Mexico	1
Portugal	1
Egypt	1

First of their nationality

Table 2: Number of nationalities that performed a private flight

As shown on the table, private space flights contributed to sending 4 new nationalities in space that have never flown there before. Private space flights also represent 16 different nationalities over the 50 that have ever been in space in total. The number of Americans is noticeable, especially since the recent flights of *Blue Origin*, *Virgin Galactic* and *SpaceX* that sent in space many Americans (some missions allowed only US citizens).

5.3. A new type of profile thanks to private flights

With the increase of space tourism, it is a common idea to hope having a more diverse range of profiles and not only military background as most of the "non private" astronauts. For instance, it could be envisaged that this new type of space flight would significantly improve diversity of profiles in general. In fact, as the ticket for

going into space with such organisations is worth more than 20 Millions of \$, only a small percentage of people can afford it. These people look like each other with their background and job. Indeed, during the first 10 years of the private space flight era, almost all of the private astronauts were entrepreneurs and/or business people, and only one of them (out of 11 launches) was a woman. It is significantly different from the first astronauts in the 1960s with their military background, but still there is a long way to go to improve diversity of profiles for both public and private space flights.

6. Analysis of the Astronaut Profile through the Perspective of Space Agencies

From the perspective of space agencies, it is important to carry out an analysis of the first 20 years of spaceflight, as well as the most recent 20 years. This type of comparative analysis helps us to understand changes in the number of astronauts, gender, ethnicity and nationality by space agencies.

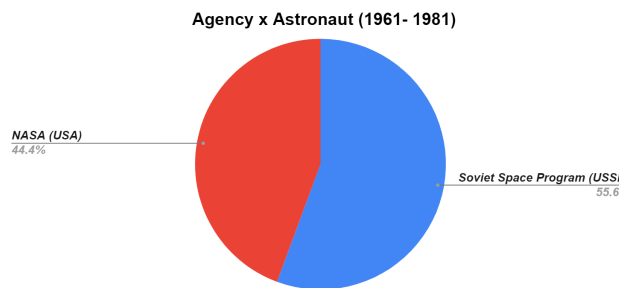


Figure 11: The number of astronauts per agency from 1961 until 1981.

The beginning of the space race and the cold war were very intense contexts in technological production. One of the biggest goals was to highlight the economic and technological power of each nation and through its space agencies, NASA and the Space Soviet Program, space missions were increasingly ambitious and innovative. In this way, it can be seen that investments were high on both sides, from equipment to training developed by highly qualified researchers, this reflection is noticed when one observes 55.6% of the astronauts from the USSR and 44.4% were from NASA, showing the fierce competition between agencies [9]

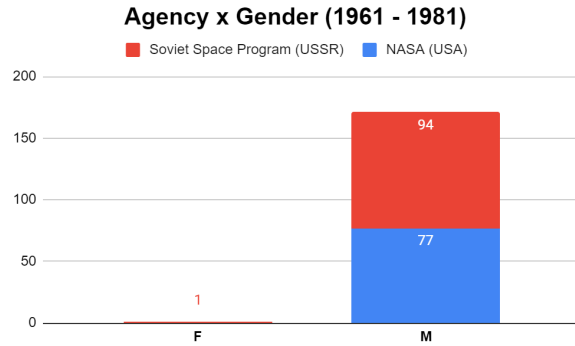


Figure 12: The number of females and males astronauts per agency from 1961 until 1981.

In the scientific and technological areas there was a strong reality of sexism, which was extended to the space sector. Aerospace gender data shows an even greater gap as, for example, in the first 20 years of human space flight, only 1 woman from the Soviet Space Program went to space. Reflecting on this context, it is noted that it was not an activity openly possible for women, even because of the restriction or difficulty of scientific contribution to the space sector. A classic example from history was the female Mercury 13 crew who performed all the necessary tests for the mission but were never allowed to go into space. [10]

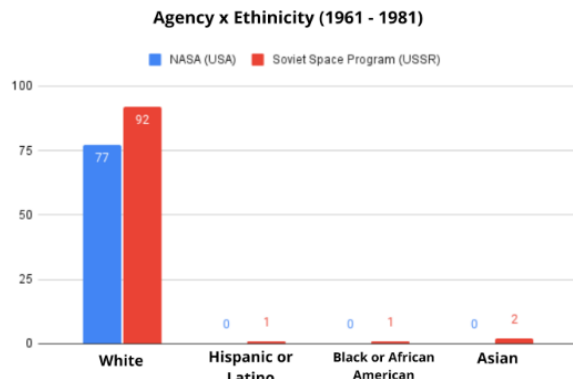


Figure 13: The ethnicity of astronauts per agency from 1961 until 1981.

The agencies were not concerned with making the space accessible to everyone, the thought at that time was to increasingly develop technology to highlight the programs on this theme in their countries. Therefore, mostly white people were astronauts and in the case of diversity, only the USSR made a slightly different reality possible, as can be seen in the graphic above. One of the factors to make it happen was due to the selection of astronauts which was targeting academics and military personnel, as explained above, also considering this audience from an advantaged social

class with more access to better and more qualified education programmes. In addition, diversity was still a huge topic in the US at that time since the Kennedy administration had sent three of the least controversial human rights treaties to the Senate for its consideration: the Supplementary Regime of Slavery Convention, the Convention on the Political Rights of Women and the Convention on the Abolition of Forced Labour. [11]

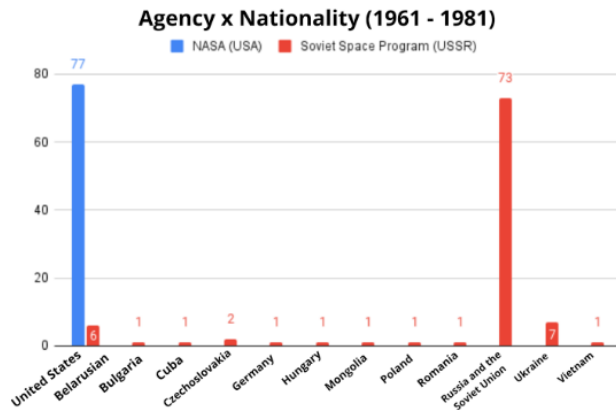


Figure 14: The nationality of astronauts per agency from 1961 until 1981.

The NASA and Soviet Space Program are structured and developed due to the context of the cold war and the space race, making it clearer that the nationalities with the largest number of astronauts are Russians and Americans. Even though the third largest nationality was Ukraine and the fourth was Belarusian, there was less diversity of nationalities because access to space was only through the two agencies of the space race. [12]

The last 20 years have been intense with technological innovations, ease of access to space, greater integration of knowledge and thus, missions are increasingly assertive, safe and comfortable. Therefore, a diversified, safe and less costly space sector is perceived. [13]

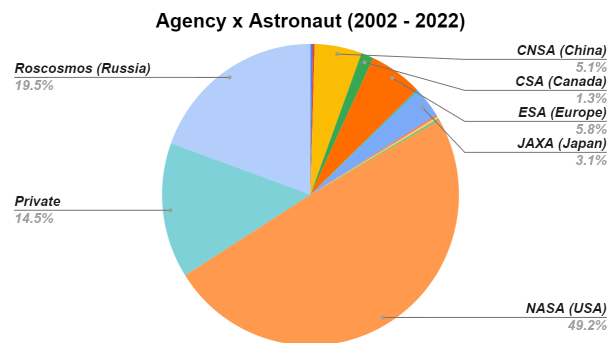


Figure 15: The number of astronauts per agency from 2002 until 2022.

There is a very big difference with the reality of 1961 - 1981, where today there is a diversity of agents active in the space sector and the growth of astronauts every year. Currently, NASA is still the agency with the most astronauts (49.2%), in second is Roscosmos (19.5%), being a vestige of the space race, having few astronauts with the completion of the USSR. Finally, a current highlight is the private ventures (14.5%) for launch and space tourism, as in a short time there was a great growth in the number of scientists or travellers. [14]

Both ESA (5.8%) and CNSA (5.1%) are agencies that are growing rapidly due to intense investments in the space area, from ambitious prolonged government plans in the area to affirmative policies in their countries with a focus on innovation and education. [15]

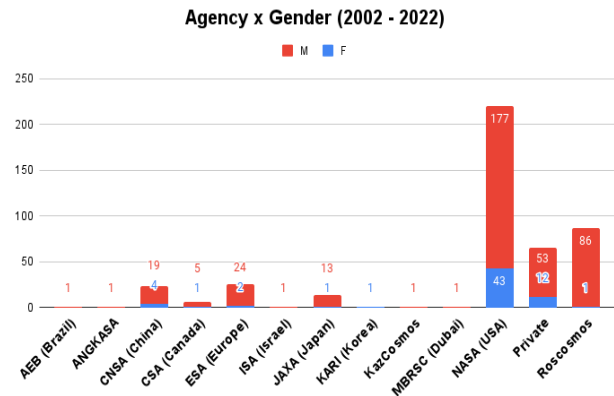


Figure 16: The number of females and males astronauts per agency from 2002 until 2022.

Another change was the current insertion of more women into the global space program. The largest number of women are from NASA, in second place is the private sector, because so far only one Russian woman has been in space. Finally, China is in third place with 4 women compared to 19 men who went to space. This integration of more women into the space sector is due both to the recent cultural shift towards women in STEAM and also to investments in affirmative policy action [16] from publicity to scientific projects including the Artemis program. This program now involves several world space agencies and aims to take the first woman and the first person of colour to the Moon which will directly impact the next space generations through the issue of representation [17].

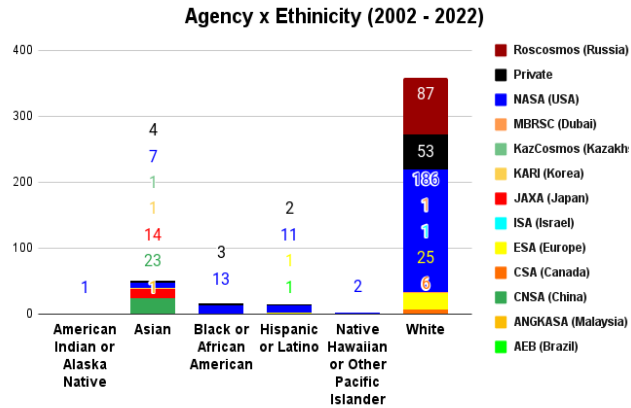


Figure 17: The ethnicity of astronauts per agency from 2002 until 2022.

Ethnicity has a profile of diversity today but without a significant amount when compared to white ethnicity since the beginning of the space race. NASA and Roscosmos are still the two agencies with the most white people, followed by the private sector and ESA in fourth place. The agency with the greatest amount of diversity is NASA, even though it has the largest number of white people, it has 34 people of different ethnicity, to the detriment of the private sector with 9 people. That's because recently, it became NASA's goal to diversify from gender, sexual orientation, to the ethnicity of astronauts. [18]

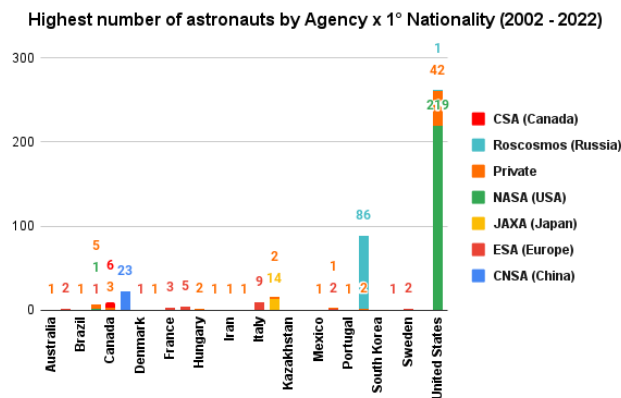


Figure 18: The first nationality of astronauts per agency with expressive human space flights from 2002 until 2022.

Diversification in the nationality of astronauts has recently increased. In the scenario of agencies with more numbers of astronauts, it is clear that due to the entire history of space, the United States is still the most expressive nationality, then Russia, the third nationality is China and the fourth is Japan. In terms of diversity, ESA and the private sector both gather the most diverse pool of nationalities, the first due to the agency

representing 22 European Member States and the second due to the absence of the nationality criterion in its selection of astronauts, as is usually present in government space agencies. [19]

In the scenario of agencies with fewer astronauts, it is noted that there are few well-established space programs until today, for example the MBRSC, KazCosmos, KARI, ISA, ANGKASA and AEB agencies with 1 astronaut each. However, even with few space flights, they present vast development in the scientific area or in projects with more consolidated agencies in complex missions. [20]

Agency x 2° Nationality (2002 - 2022)

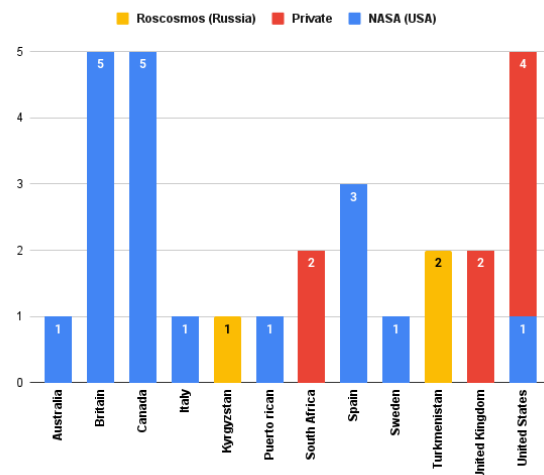


Figure 19: The second nationality of astronauts per agency from 2002 until 2022.

In terms of dual nationality, around 4.48% of US astronauts have another nationality, which was accounted for in the graph. Finally, it can be seen that the British and Canadians have the most percentage of astronauts with the second nationality and then Spain with 3 astronauts. It is also noted that in terms of second nationality, NASA is the agency with the highest number of such cases (17 astronauts), followed by the private sector with 8 astronauts.

7. Analysis of the Astronaut Profile through the Perspective of Gender from 1961 to 2022

For many years in history the science and technology industry, or in this case specifically the space sector has been predominated by male. Women were not part of the Human spaceflight programs, both for the American and Russians. Three years after the first human who travelled to space was Yuri Gagarin, a male cosmonaut, then the first female astronaut was sent to space. In 1963 Valentina Tereshkova, a Soviet cosmonaut became

the first female to travel to space when she was launched on a solo mission aboard the spacecraft Vostok 6. Based on the analysis done for this project, it was also important to understand and to showcase the comparison on gender participation in each space mission throughout the years.

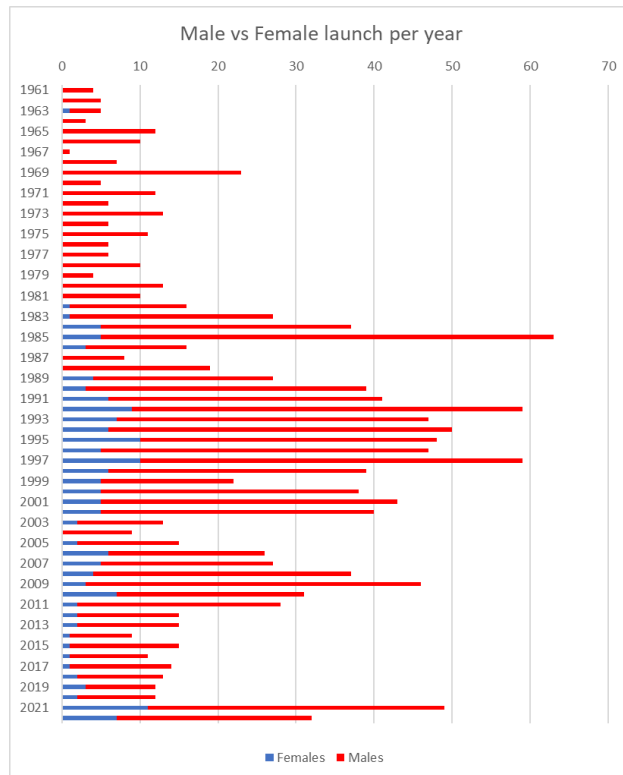


Figure 20: Comparison of male and female participation in space missions from 1961 to 2022

Valentina Tereshkova is the 12th astronaut, after 11 American and Russian men. After her, humanity has had to wait for 19 years before seeing another woman flying in space: Svetlana Savitskaya. In terms of numbers, 171 male astronauts flew between the 2 first women. A short time after, American astronaut Sally Ride made the first US female flight.

The reason behind this seems to be linked to a lack of knowledge on womens' bodies, combined with a societal idea that women were too "weak" for this harsh environment and would need to be preserved. In the 1960s and 1970s, space activity was limited to the Cold War between the US and the Soviet Union. It thus took place in times where most women stayed home while their husbands served in the 2 previous world wars.

Also, in 1981 was the beginning of the US Space Shuttle program and thus allowed more varied profiles to go to space, thanks to bigger crews. We could see for the first time some people with unusual backgrounds,

scientists, teachers, docteurs, etc. Less military means also less astronauts coming from a very male environment, and thus more chances to see women doing space flights.

In terms of percentages, there have been only 11,9% of women space flights since the first astronaut launched in space, but since the beginning of the Shuttle, the number has risen to 13,5%.

Since the recent increase of space tourism activities, diversity has increased as well once again thanks to the fact that even more several backgrounds, jobs, hobbies were accepted to go in space. Since 2017, females represent around 20% of the spaceflights. With the other private launches like SpaceX, Blue Origin and Virgin Galactic, there are even easier ways to reach that limit of sky for again more diverse profiles. Indeed, we can see in this context the number of 22% representing females. In the end, the recent number could let us imagine we will soon reach a strong quarter of female flights, which is still low compared to their male colleagues.

8. Analysis of the Astronaut Profile Gender through the Perspective of Countries

Just as previously mentioned, the space race, a 20th Century struggle between two nation-states, the Soviet Union (USSR) and the United States (US) had the intention of pursuing domination of space flight technologies. The Soviet Union decided to send a woman into space merely to score propaganda points against the U.S. by being the country who chose 5 women for a cosmonaut training program and then send the first female to space. Although this was for propaganda purposes, it surely inspired many women, girls and even men. Followed by cosmonaut Valentina Tereshkova, the Soviet Union trained Svetlana Savitskaya and launched her on a mission to the Salyut 7 space station thus the Soviet Union made sure that the first two women in space were Soviet citizens. Ever since then only 2 more women cosmonauts from Russia or the Soviet Union have flown to space and it is expected that in October 2022 Anna Kikina becomes the the fifth female cosmonaut to fly to space and the first Russian to travel aboard a Crew Dragon spacecraft.

Referring to the graph below we can see that from the 42 countries that have an astronaut or many astronauts, the US is the country with the most astronauts who have flown to space with a percentage of 59.71%. Additionally, the US is the country that has sent more female astronauts to space with about 76.41%. Followed by Russia or the Soviet Union with a percentage of 24.9% of both female and male astronauts.

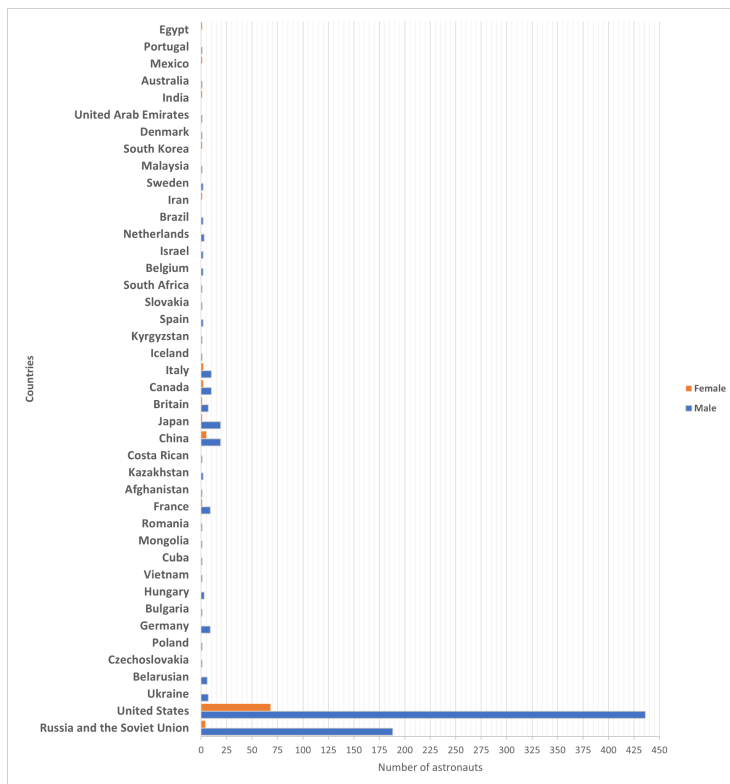


Figure 21: Comparison of male and female participation in space missions by countries

After the US and Russia or the Soviet Union we have France, China, Japan, Canada, Italy, Iran, South Korea, India and Egypt with at least one female astronaut. In fact, from some of these countries their first astronauts were females such as Iran, South Korea, India and Egypt. Just as mentioned in our previous analysis, private companies and the very recent space tourism has helped a lot in increasing the numbers of female astronauts and also having the collaboration between female and male astronauts towards a more inclusive space sector, specifically astronauts.

9. Conclusions

The construction of the profiling of selected astronauts who went to space from 1960 to the present day is crucial for understanding the evolution of the space scenario in professional, governmental, ethnic, geographic and inclusive aspects.

The profiles were most successfully collected in the first 20 years of the space race and the most recent 20 years, which shows the great diversity of numbers in terms of public and private cooperation, more women astronauts, different professional backgrounds, other nations entering in the space sector, changing the types and

period of missions, as well as noting that there is still a lot to be done even with these recent changes.

We can compare historical data to predict trends in upcoming crewed space missions but, perhaps more importantly, we must consider the context of each mission to understand how technology readiness, budget and schedule constraints, political motivation, and scientific objectives influenced crew selection decisions.

An interesting possibility of these types of work are databases for comparing trends in upcoming human space missions. That's because it is important to look at the context of each mission to understand how technology readiness, budget and schedule constraints, political motivation, worldwide cooperation, and scientific objectives influence crew selection decisions.

For future work, it is necessary to finalise the data collection to have a complete profile between 1961 and 2022 in order to complete a more robust analysis. Another fascinating future work would be to investigate statistics on crew launches during the Artemis Program and, eventually, voyages to Mars to understand the profile of past expeditions and conquests.

Acknowledgements

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References

- [1] Logsdon, John M. Exploring the unknown: selected documents in the history of the US civilian space program. Accessing space. Vol. 4. NASA, 1999.
- [2] Siddiqi, Asif A. "The Almaz Space Station Complex-A History 1964-1992-Part 2 1976-1992." Journal of the British Interplanetary Society 55 (2003): 35-67.
- [3] Compton, W. David, and Benson, Charles D. Living and Working in Space: A History of Skylab (NASA SP-4208, 1983)
- [4] Alexander, C. C.; Grimwood, J. M.; Swenson, L. S. (1966). This New Ocean: a History of Project Mercury. US: NASA. ISBN 1934941875.
- [5] Kitmacher, G. H. (2006). Reference guide to the international space station (No. NASA/SP-2006-557).

- [6] US Office of Management and Budget, Revisions to the Standards for the Classification of Federal Data on Race and Ethnicity, 1997, <https://www.govinfo.gov/content/pkg/FR-1997-10-30/pdf/97-28653.pdf>
- [7] B. Dunbar, 60 Years and Counting - Human Spaceflight, <https://www.nasa.gov/specials/60counting/spaceflight.html#:~:text=The%20objectives%20of%20the%20program,the%20astronaut%20and%20spacecraft%20safely>.
- [8] D. Harland, "Skylab | History, Discoveries, & Facts," *Encyclopedia Britannica*. <https://www.britannica.com/topic/Skylab>
- [9] Cross, M. K. D. (2019). The social construction of the space race: then and now. *International Affairs*. doi:10.1093/ia/iiz190
- [10] Wade, Jessica. "Thirteen tales that must be told". *Physics World* 31.12 (2018): 40
- [11] Bradley, Curtis A. "The United States and Human Rights Treaties: Race Relations, the Cold War, and Constitutionalism." *Chinese Journal of International Law* 9.2 (2010): 321-344.
- [12] Starr, Kristen Amanda. *NASA's hidden power: NACA/NASA public relations and the Cold War, 1945–1967*. Auburn University, 2008.
- [13] Paikowsky, D. (2017). What Is New Space? The Changing Ecosystem of Global Space Activity. *New Space*, 5(2), 84–88. doi:10.1089/space.2016.0027
- [14] Sehovic, Irhad. The NewSpace race: the private space industry and its effect on public support for NASA funding. working paper, March 2018, 52 p, available at https://www.gym.edu/36133518/The_NewSpace_Race_The_Private_Space_Industry_and_Its_Effect_on_Public_Support_for_NASA_Funding, 2018
- [15] Ribeiro G., Global space economy maintains growth in 2020 and reaches US\$ 371 billion, <https://mundogeo.com/2021/08/11/economia-espacial-global-mantem-crescimento-em-2020-e-atinge-us-371-billions/>
- [16] UNOOSA. History and Background of Space4Women. March, 2022. <https://space4women.unoosa.org/content/historyand-background-space4women>. (accessed 08.23.22).
- [17] NASA, Artemis, <https://www.nasa.gov/specials/artemis/>
- [18] Wong, Sharon. "Diversity—Making space for everyone at NASA/Goddard Space Flight Center using dialogue to break through barriers." *Human Resource Management: Published in Cooperation with the School of Business Administration, The University of Michigan and in alliance with the Society of Human Resources Management* 47.2 (2008): 389-399.
- [19] Górecki I., Wanting to become an astronaut: How astronaut selection has changed over 60 years of human space exploration, <https://bmsis.org/wanting-to-become-an-astronaut-how-astronaut-selection-has-changed-over-60-years-of-human-space-exploration/>
- [20] Pankova, Ludmila V., Olga V. Gusarova, and Dmitry V. Stefanovich. "International Cooperation in space activities amid Great Power Competition." *International Cooperation* (2021).